

摘要：

相比较5月25日发布的V1版本，新版论文在原有理论框架基础上，补充了大量工程落地细节、实测量化数据与产品演进路线，进一步完善了以时间常数 τ 为核心的后摩尔时代缩放理论体系。

根据中国科学院科技论文预发布平台 ChinaXiv 最新公示论文，华为半导体负责人何庭波于 7 月 3 日发布《面向多层级电子系统的时间缩微理论》（业内也称“**稻定律**”）V2 版本。

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在论文结构方面，V2 版整合 V1 版引导段落，形成 8 章完整论述体系，章节逻辑分层更清晰。

V2 版还新增了多张原理与实物示意图，覆盖 τ 分层时空模型、LogicFolding 架构、键合界面截面、Unified Bus 互连架构、Hi-ONE 光引擎等核心技术。

在工程落地方面，V2 版本深度阐释核心技术 LogicFolding 的齿比（gear ratio）概念，在混合键合间距接近顶层金属布线尺寸时，3D 设计空间从传统的“宏块级离散优化”转向“单元级连续优化”，可实现全局最优的垂直逻辑划分，突破了传统 3D 堆叠仅能按功能块分层的局限。

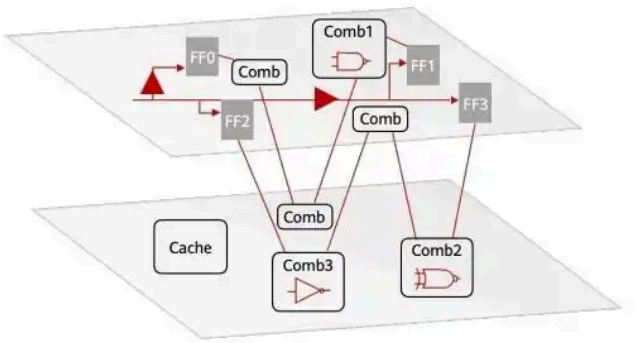


Figure 2. The schematic illustration of *LogicFolding*.



V2 版还新增量产实测数据表，明确给出 Kirin 2026 与基准 Kirin9030 Pro 的电压、频率、归一化功耗、面积与功率密度参数。

Table 1. The comparison of power at iso-performance between Kirin 2026 and Kirin9030 pro

	Kirin9030 Pro	Kirin2026
Temperature	25 °C	25 °C
Voltage	1.1 V	0.9 V
Frequency	2.75 GHz	2.5 GHz
Normalized power at iso-performance	1	0.59
Normalized area	1	0.625
Normalized power density	1	0.944

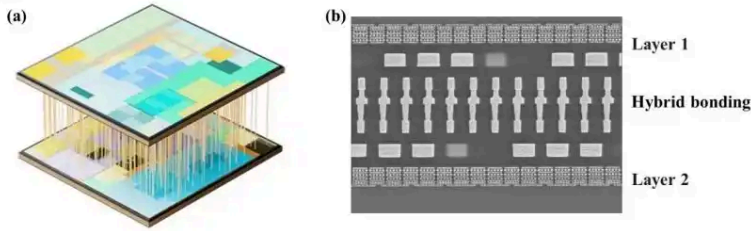


Figure 3. (a) The schematic illustration of the next-generation Kirin SoC platform; (b) Cross-section image of its bonding interface.

Table 2. Trend of the operating frequency of Kirin CPU performance core.

	SoC	Architecture	Frequency	State
2023	Kirin9000s	Planar	2.6 GHz	Mass product
2024	Kirin9020	Planar	2.65 GHz	Mass product
2025	Kirin9030 pro	Planar	2.75 GHz	Mass product
2026	Kirin 2026	LogicFolding	3.1 GHz	Silicon
2027	Kirin 2027	LogicFolding	3.39 GHz	Silicon
2028	Kirin 2028	LogicFolding	3.71 GHz	Pre-silicon
2029	Kirin 2029	LogicFolding	4 GHz	Pre-silicon

V2 版还细化全场景路线图，明确技术演进节点，在移动端补充 TSV 从顶层金属下移至 M6 层、多有源层堆叠等演进路径；在 AI 端明确 Ascend 系列加速器的迭代节奏。

The roadmap places this evolution on an explicit timeline. Through approximately 2030, AI accelerators (the Ascend SuperPoD line — Ascend 910C in 2025, Ascend 950 in 2026, and the 990to follow) rely on a combination of mature techniques: chiplets, 2.5D fan-out, and 3D stacking via micro-bump and standard-pitch hybrid bonding. Around 2030, Ascend 990 will introduce *LogicFolding* into the AI accelerator class, and from that point 3D Folding becomes the principal carrier of α through 2035. Along this path, hardware integration is projected to increase by more than $100\times$ by 2035, with τ reduction distributed across every layer of the stack rather than concentrated at the device level.

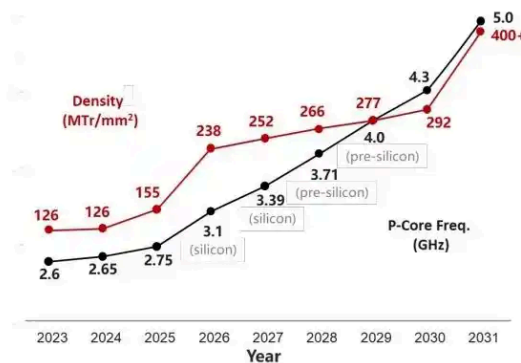


Figure 4. Projected transistor density and performance-core frequency for our future Kirin products.

Highlight — τ at AI system scale



- UB remote-access latency: ~ 10 s of μ s $\rightarrow \sim 100$ ns ($\approx 500\times \tau$ reduction)
- Hi-ONE per-module bandwidth: 8 Tb/s (matches per-chip UB bandwidth)

This version posted 2026-07-03.

- Hi-ONE SerDes reach: ~ 100 cm $\rightarrow \sim 5$ cm; panel-to-panel reach: < 1 m $\rightarrow 100$ m
- Fan-out dilemma: compute $\propto N^2$, perimeter-bound BW/I/O/power $\propto N$
- 3D Folding: relocates BW, optical I/O, and power delivery from edges onto surfaces, restoring N^2 parity
- 2026 $\rightarrow$ 2035 projected hardware-integration growth: $> 100\times$

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